

additions_synthesis

Helix OTA — Synthesis & Reconciliation of additions/ Input Drafts

Field	Value
Revision	3
Created	2026-06-07
Last modified	2026-06-08
Operator decision (2026-06-08)	End-user rollback is in 1.0.1 (with staged-rollout), not 1.0.2 — §10 K8 + §11 updated. G3/G4/G5 operational endpoints + the staged-rollout engine + migration 002 are now IMPLEMENTED and real-DB/real-brick verified (see §8.1).
Status	active
Status summary	Deep analysis of all three operator-supplied inputs in docs/research/main_specs/additions/ — initial_research_01.md, initial_research_02.md, and the large initial_research_03/helix_ota_big/corpus (26 markdown design docs: SERVER_IMPLEMENTATION, ANDROID_CLIENT_DESIGN, REST_API_SPECIFICATION, VERSION_ROADMAP, ROLLBACK/DELTA/LINUX/WINDOWS/MULTI-OS research, CODE_REVIEW_AND_GAP_ANALYSIS). Extracts every reusable element, maps each to a phase/spec/code destination, and records every contradiction (between inputs, against locked decisions, against the Constitution) with a resolution rule. This document is the traceability bridge between operator input and the canonical specs. Per-input exhaustive inventories live in additions_analysis/0{1,2,3}_analysis.md; §8–§12 below consolidate them.

Field	Value
Issues	Inputs conflict on wrapped-engine (Mender vs hawkBit), topology (modular vs microservices), router (chi vs Gin), signing (RSA vs ed25519), transport auth (mTLS vs JWT), TUF-now vs deferred, and the staged-rollout-in-MVP question. All resolved in §5 + §10 in favor of the locked decisions / ADRs.
Issues summary	No input conclusion is treated as settled where it conflicts with a locked decision or an evidence-based ADR. The locked decisions win; divergent input choices are recorded and routed.
Fixed	Rev 2: processed addition #3 (the helix_ota_big corpus) + re-processed #01 (incl. its 2026-06-08 update); added the consolidated 3-input gap register (§8), conflict reconciliation (§10), future-phase catalogue (§11), and the new-work-item routing (§9). Rev 1: drafts 01+02 synthesized.
Fixed summary	All three additions processed per operator mandate (additions = authoritative input; nothing skipped or assumed). Every requirement is routed to a COVERED spec/ code location or to a tracked gap with a destination.
Continuation	Implement the routed gaps in §8/§9 (MVP-critical first); write the staged-rollout engine spec (1.0.1) and the dashboard spec; fold the addition-#3 future-phase research into the versioned dirs per §11. Re-run this synthesis whenever a new additions/ file lands.

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§1. Purpose & method

Per operator mandate, every file in additions/ is authoritative input that MUST be deeply analyzed and used. This document performs that analysis for the two drafts present, applying three lenses:

1. **Extract** — what concrete, reusable material does the draft contribute (APIs, schemas, diagrams, code, sequencing)?
2. **Reconcile** — where do the drafts disagree with each other, with the locked decisions, or with the HelixConstitution?
3. **Route** — where does each accepted element land in the canonical corpus, and which contradictions are deferred to an evidence-based Architecture Decision Record (ADR)?

Governing constraints: the locked decisions (Native A/B + custom Go control plane; foundation-corpus-first; research decides the wrapped engine; pre-authorized public repo creation), the mandated stack (Go + Gin + Brotli + HTTP/3(QUIC)→HTTP/2 fallback, REST primary), and the HelixConstitution (§11.4.6 no-guessing, §11.4.8 research-before-implementation, §11.4.74 catalogue-first reuse, §1 four-layer testing).

§2. Source inventory

Draft	Core thesis	Strongest contributions
initial_research.md	Wrap Mender as the core; Go server orchestrates it; Android UpdateEngine client; per-phase directory tree.	OTA comparison matrix; /api/v1 endpoint sketch; first-cut PostgreSQL schema (releases, telemetry_events, rollback_history); update sequence + ER + container + staged-rollout Mermaid diagrams; per-phase directory layout; multi-format export plan.
initial_research_02.md	Wrap Eclipse hawkBit + adopt TUF for supply-chain security; microservices topology; pluggable OS-adapter architecture.	Android 15 Virtual A/B + compression deep dive; OTA .zip / payload.bin structure; update_engine.applyPayload properties; richer DB schema (users, devices, groups, artifacts, deployments, phases, device_deployments, rollouts, audit_logs, update_metrics + indexes); REST + gRPC surface; phased-rollout JSON + Go rollout-engine logic; go-tuf/v2 signing/verify sketch; docker-compose + k8s manifests; Kotlin client + foreground service; OS-adapter interface; 48-week roadmap.

§3. Points of agreement (high-confidence inputs)

Both drafts independently converge on these — treated as high-confidence (still validated, never assumed):

- **Device-side = Android A/B via update_engine** with automatic boot-failure rollback (matches our locked strategy and the AOSP anti-corruption guarantee). Draft 02 correctly refines this to **Virtual A/B with compression** for Android 15.
- **Server stack = Go**, PostgreSQL for relational state, **MinIO/S3** for artifact blobs.
- **Artifact integrity = SHA-256 + signature verification**, verified server-side on upload and device-side before apply.
- **Staged rollout** with percentage phases (5/10/30/.../100) gated by success/error thresholds, with pause/halt on breach.
- **Telemetry** event stream from devices feeding dashboards.
- **Containerized** deployment; Constitution-governed; multi-format documentation export.

§4. Extracted reusable elements → destination

Element (source)	Accepted?	Destination in corpus
OTA solution comparison matrix (01 §2.1; 02 §1)	Accept as input, expand with evidence	research/ ota_landscape_report.md + ADR-0001 (engine selection)
Android 15 Virtual A/B + payload.bin/.zip structure, applyPayload properties (02 §2)	Accept (factual, verify against AOSP docs)	1.0.0-mvp/client_android/ update_engine_integration.md
/api/v1 REST endpoints (01 §5.1; 02 §5.1)	Accept, merge, re-base on Gin + REST-primary	1.0.0-mvp/api/openapi.yaml + endpoints.md
gRPC service defs (02 §5.2)	Accept as optional/internal only	1.0.0-mvp/api/ internal_grpc.md (REST is the mandated surface)
PostgreSQL schemas (01 §5.2; 02 §4)	Accept 02 as base (richer), merge 01, normalize	1.0.0-mvp/database/ (migrations + schema.md)
Phased-rollout JSON + Go rollout-engine loop (02 §6)	Accept as design seed for the rollout engine	1.0.1-staged-rollout/ rollout_engine.md (engine is OS-agnostic submodule)
go-tuf/v2 signing/verify sketch (02 §7.1)	Accept as the trust-model candidate	research/ supply_chain_trust_adr.md (ADR-0002 Uptane/TUF), security/ specs
Kotlin client + foreground service (02 §10)	Accept as reference; rework to KMP +	1.0.0-mvp/client_android/ code snippets

Element (source)	Accepted?	Destination in corpus
	WorkManager + jitter	
OS-adapter interface (02 §11.3)	Accept as the universality seam	00-master/2026-06-07-helix-ota-design.md (§4 architecture) + future-OS phase dirs
docker-compose + k8s manifests (02 §9)	Accept, re-base on the containers submodule	1.0.0-mvp/deployment/
Mermaid diagrams (01 §3.1/§8)	Accept	1.0.0-mvp/architecture/diagrams/*.mmd
Per-phase directory tree + export plan (01 §7/§12)	Accept, re-root under docs/research/main_specs/	00-master/ + export pipeline spec

§5. Contradictions & resolutions

#	Contradiction	Resolution
C1	Mender (01) vs hawkBit (02) as wrapped core.	Neither is settled. Both enter ADR-0001 with AOSP-native-only as a third option. The evidence-based research report scores them; the locked strategy biases toward <i>minimal wrapping</i> (native A/B + custom Go control plane), so a full-platform adopt must clear a high bar.
C2	Microservices (02) vs a leaner modular control plane.	MVP favors a modular monolith (one deployable, internal package boundaries mirroring the future services) to cut operational cost; the OS-adapter + rollout-engine seams stay extractable. Microservices revisited at scale (ADR-0003 topology).
C3	Poll cadence: 6 h (02 service) vs X-min (01).	Locked default 15 min + jitter, configurable (operator decision). Drafts' constants are non-binding.

#	Contradiction	Resolution
C4	API surface: REST+gRPC (both) vs mandated Gin + REST primary, HTTP/3→HTTP/2 fallback, Brotli.	Mandated stack wins. REST is the compatibility surface; gRPC is optional/internal only. All endpoint sketches re-based onto Gin + the http3 submodule.
C5	TUF/Uptane timing: 02 puts TUF in MVP security service; locked scope defers full Uptane to 1.0.1+.	MVP ships signed-artifact + SHA-256 + AVB; TUF/Uptane is an ADR-0002 hardening item for 1.0.1+. Signing interfaces are designed MVP-forward so Uptane drops in without rework.
C6	Rollback scope: 02 roadmaps user-rollback/multi-version early; locked scope defers end-user rollback past MVP.	MVP = automatic A/B boot-failure rollback only. End-user/multi-version rollback = 1.0.1+ (matches operator “not for first version”).
C7	Redis (02) as session/cache store.	Optional; prefer the cache submodule abstraction. Adopt Redis only if the catalogue brick doesn’t meet need (§11.4.74).
C8	Coverage target 80% (02) vs 90% (01).	Constitution §1 four-layer coverage + mutation immunity supersedes any single percentage; numeric target set per-component, floor ≥90% for safety-critical paths (signing, apply, rollout-gate).

§6. Corrections of factual/guessed claims

- **Guessed submodules** in 01 §6.2 / 02 §13 (go-common, helm-charts, vasic-digital/secrets) **do not exist** under those names. Use the verified catalogue: auth, security, database, Storage, observability, eventbus, ratelimiter, middleware, http3, mdns, recovery, Herald, config, discovery, cache, docs_chain/Document/Formatters, and KMP Auth-KMP/Security-KMP/Storage-KMP/Config-KMP. Secrets management → evaluate security + container secrets before introducing Vault.
- **hawkBit stats** in 02 §1.1 (“579 stars ... last commit June 5 2026”) are unverified specifics — must be re-checked live in the research report; do not propagate as fact (§11.4.6).

- “**Mender client is C++**” (02) and ***“Android?”*** (01) — Mender’s Android support is weak/unofficial; this materially weakens C1’s Mender option and must be confirmed in ADR-0001.
- **file:// payload to applyPayload** (02 §10) vs **HTTPS streaming** (02 §2.2) are two different apply paths; the spec must pick and justify (local-verified-file apply is safer for our verify-before-apply requirement).
- Draft 02’s self-description as “simulated analysis” / “most comprehensive ever” is marketing, not evidence; stripped from canonical docs per §7.1 no-bluff.

§7. Open questions routed to research ADRs

- **ADR-0001** Wrapped engine: AOSP-native-only vs hawkBit vs Mender (resolves C1).
- **ADR-0002** Supply-chain trust: plain signing vs TUF vs full Uptane, and MVP-vs-1.0.1 timing (resolves C5).
- **ADR-0003** Server topology: modular monolith vs microservices, and the scale trigger (resolves C2).
- **ADR-0004** Transport: HTTP/3(QUIC)+Brotli rollout, HTTP/2 fallback negotiation, device-network realities (anchors C4).
- **ADR-0005** Delta updates: AOSP block diffs vs custom, and phase placement.

§8. Consolidated gap register (all three inputs)

Deduped union of every MISSING/PARTIAL requirement surfaced by the three per-input analyses (additions_analysis/01_analysis.md, 02_analysis.md, 03_analysis.md). Severity: **C** = MVP-critical (code-actionable now), **H** = high (next phase), **M** = medium (catalogue/defer). Each row routes to a destination — nothing is dropped.

ID	Gap (source)	Sev	Status vs built system	Destination / action
G1	Anti-downgrade invariant on the update-check offer path (03 R16; 01)	C	MOSTLY COVERED — handleClientUpdate returns 204 when current ≥ release (so a strictly-lower target is never offered for a known, parseable version); release-creation enforces S4 monotonicity. Residual: no regression test pinning the invariant; an empty/unparseable current falls through to “offer”. UNVERIFIED at spec level — ordering lives in ota-artifact-validator; handler maps S2→422 HASH_MISMATCH	Add regression tests proving downgrades/equal are never offered + behavior on empty/unparseable current; document the invariant in endpoints.md §12.1. Implemented this pass — see §8.1. Confirm S2-before-S3 in the validator brick + state it explicitly in security/signing_verification.
G2	Validation-chain order hash-before-signature is verified+documented (03 R15)	C		

ID	Gap (source)	Sev	Status vs built system	Destination / action
			before S3→422 SIGNATURE_INVALID.	
G3	Audit logging (audit_logs table + write path) (01 REQ-01-60; 02; 03 R-audit)	H	PARTIAL — table planned in database/schema.md; no audit middleware in server/internal/api.	Spec an audit middleware - AppendAudit repo method 1.0.1 (compliance).
G4	Telemetry read/aggregation endpoints — per-device history + /telemetry/overview (02 R02-5.8; 03 R26)	H	PARTIAL — ingest built (POST /client/telemetry, schema-validated); no read side.	Spec read endpoints in endpoints.md; dashboard monitoring phase.
G5	Device-group CRUD + membership endpoints (02 R02-5.7; 03 R20)	H	PARTIAL — group accepted in DeploymentCreate; group tables in schema; no CRUD routes.	Spec group endpoints (1.0. or MVP if grouped deploys needed sooner).
G6	React dashboard (upload→deploy→fleet/telemetry view) (01 REQ-01-78; 02; 03 R23)	H	MISSING — no spec, repo, or code anywhere. Largest single doc gap.	New dashboard/ spec reusing UI-Components-React/Dashboard-Analytics-React/Auth-Context-React; new public repo.
G7	Full staged-rollout engine: phase schema + SQL + Go engine + lifecycle ops (start/pause/resume/abort/rollback) (02 R02-6.x; 03; 01)	H	PARTIAL — design seed only (master §8, 1.0.1-staged-rollout/README.md); ota-rollout-engine brick exists (halt-wins + deterministic cohorts).	Promote 1.0.1-staged-rollout outline → full spec + migration 002_* (deployment_phases, rollouts, rollback_history). Biggest unwritten spec body.
G8	Mandatory-update / deadline semantics (03 R18)	M	MISSING — mandatory/deadline undefined everywhere.	Define in update-check contract; 1.0.1.
G9	Tamper event + rollout-halt-on-HASH_MISMATCH (SECURITY_TAMPER_DETECTED) (03 R27)	M	MISSING.	Add to telemetry-schema + rollout-gate; 1.0.1 security hardening.
G10	CI / governance: GitHub Actions, CODEOWNERS, gosec, Dependabot (01 REQ-01-75)	M	MISSING from repo.	Add CI workflows running all validators (redocly, migrations, kubeconform, gtest, gradle :core:test, expo render).
G11	NEW ota-* submodule repos created PUBLIC on GitHub+GitLab (01 REQ-01-62/47)	M	UNVERIFIED — submodules wired locally; repo existence per-host not confirmed in this pass.	Verify/create per host; mirror to GitFlic/GitVerse (§4).
G12	NFR targets bound + load harness: 10k devices, 99.9%	M	UNVERIFIED — asserted, never measured.	Build a load harness; set numbers from measured MVP load (do not assert).

ID	Gap (source)	Sev	Status vs built system	Destination / action
	SLA, <100ms (02 R02-8.4/14.2/14.3)			
G13	Secrets management wired (container-native + security/config), Vault rejected (01 REQ-01-55)	M	PARTIAL — path chosen, not wired/verified.	Wire + verify in deployment spec.
G14	mTLS device auth (02; 03)	M	DEFERRED by design — JWT bearer for MVP.	1.0.1+ hardening; keep signing-forward so it drops in.

§8.1 Implemented this pass

- **G1** — added update-check regression tests in `server/internal/api` proving the anti-downgrade invariant (a strictly-lower or equal target version is never offered; empty/unparseable reported version behavior pinned), and documented the invariant. These also raise `handleClientUpdate` branch coverage toward the $\geq 90\%$ safety-critical floor.

§9. New work items routed to specs/phases

These are net-new deliverables the inputs require that are NOT yet in the canonical roadmap as concrete specs. Each is routed; none are MVP blockers except where marked.

1. **Dashboard spec + repo** (G6) — `dashboard/` + new public repo; reuse `*-React bricks`.
2. **Staged-rollout engine full spec** (G7) — `1.0.1-staged-rollout/` promotion + migration `002_*`.
3. **Audit subsystem** (G3) — `middleware + audit_logs` write path; `security/ + database/`.
4. **Telemetry read/analytics API** (G4) — `endpoints.md` + aggregation queries.
5. **Device-group CRUD** (G5) — `endpoints.md` + repo methods.
6. **CI/governance** (G10) — `.github/workflows/`, CODEOWNERS, gosec, Dependabot.
7. **Load/NFR harness** (G12) — `tests/load/`; bind NFR numbers from measurement.

§10. Addition #3 conflict reconciliation (locked side wins)

Addition #3 is a **parallel, internally-divergent generation** (its own `CODE_REVIEW_AND_GAP_ANALYSIS.md` audits its docs against *each other*, not against the built server). Re-scored against the actual server/ build and locked decisions:

#	Addition-#3 position	vs	Resolution
K1	chi router	LOCKED Gin (built)	Gin wins.
K2	RSA-4096 / RSA-PSS signing (+ Vault type=rsa-4096)	built ed25519 + SHA-256 / ADR-0002	ed25519 canonical.
K3	mTLS device transport	built JWT bearer	JWT for MVP; mTLS optional hardening.
K4	Module paths vasic-digital/helix-ota-server, dev.helix.ota.* (self-conflicting)	built github.com/HelixDevelopment/helix_ota	HelixDevelopment/helix_ota .
K5	Vocabulary updates+rollouts	canonical releases+deployments	releases/deployments ; staged rollouts → 1.0.1.
K6	Full staged-rollout engine in MVP	locked all-targets-only MVP	engine = 1.0.1 design seed.
K7	Coverage 85%	≥90% safety-critical floor (§1/C8)	≥ 90% floor on safety paths.
K8	Numbering: rollback@1.0.1, delta@1.0.2	operator decision (2026-06-08)	end-user rollback is in 1.0.1 (with staged-rollout) — backed by migration 002's rollback_history table (real-DB validated). 1.0.2 slot is now folded into 1.0.1 (the 1.0.2-rollback/ dir is marked superseded, not deleted, per §11.4.124). delta stays 1.0.3 (aligns ADR-0005).

Addition-#3 “CRITICAL” findings that are NOT real against the build (artifacts of its own multi-author generation): endpoint-path drift (C1/C3/C4), enum drift (C5/C6/C8), os_type CHECK (C7), module path (C10), UUID-vs-prefixed-IDs (C11), registration body (C2). The built server already settled canonical paths (POST /devices/register, GET /client/update, POST /artifacts/upload), single-source enums from ota-protocol/ota-telemetry-schema, opaque string IDs, and the canonical module path.

§11. Future-phase research catalogue → versioned dirs

Addition #3 ships substantial future-phase research (markdown). Routed to the versioned phase dirs; numbering reconciled per K8 (staged-rollout owns 1.0.1):

Addition-#3 doc	Destination dir	Notes
ROLLBACK_DESIGN.md (server-triggered + user-initiated + multi-version)	1.0.1-staged-rollout/	Operator decision 2026-06-08: end-user rollback moved INTO 1.0.1. Auto boot-failure rollback is already MVP; this is the <i>operator/user-initiated</i> superset, backed by migration 002 rollback_history. 1.0.2-rollback/ is kept as a superseded pointer.
DELTA_UPDATES_DESIGN.md	1.0.3-delta-updates/	Aligns ADR-0005 (AOSP block diffs vs custom).
LINUX_OTA_RESEARCH.md (RAUC/SWUupdate/Mender)	1.X-linux/	Per-OS adapter behind the universality seam.
WINDOWS_OTA_RESEARCH.md	1.X-windows/	Per-OS adapter.
MULTI_OS_UNIVERSAL_DESIGN.md	1.X-other-os/ (or 2.0.0-multi-os-universal/)	The universal OS-adapter platform.
OTA_SYSTEMS_RESEARCH.md	merge into research/ota_landscape_report.md	Dedupe against existing landscape report.
8 diagrams	00-master/diagrams/	Dedupe against existing diagram set.

§12. Anti-bluff / UNVERIFIED register

Carried forward from the per-input analyses; these **MUST NOT** be propagated as fact (Constitution §11.4.6 no-guessing, §7.1 no-bluff):

- hawkBit/Mender repo stats and “Mender client is C++/weak Android” — confirm live in ADR-0001.
- Invented submodules (go-common, helm-charts, vasic-digital/secrets, vasic-digital/auth) do **not** exist under those names — use the verified catalogue.
- ota-artifact-validator exact S2-before-S3 ordering — verify in the brick (G2).

- AOSP `payload_properties` optional-header set on Android 15 / RK3588 — UNVERIFIED hardware gate.
- Storage brick Range-GET / pre-signed-URL behavior — UNVERIFIED until inspected.
- Addition-#3 delta-update benchmarks and NFR numbers (10k devices, 99.9%, <100ms) — UNVERIFIED, never measured.
- All HelixConstitution §11.4.x clause numbers cited corpus-wide remain UNVERIFIED except the six confirmed in `tests/test_strategy.md` §13.